

ABERPORTH BUOY 62301, United Kingdom

WMO: 996850

Lat: 52.300N

Lon: 4.500W

Elev: 0

StdP: 101.33

Time Zone: 0.00 (EUW)

Period: 01-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.0	2.9	-4.1	2.7	4.6	-2.7	3.0	3.8	17.5	9.0	16.0	8.9	6.8	70	0.730

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.7	18.7	16.7	17.9	16.2	17.3	15.8	17.1	18.0	16.6	17.5	16.1	17.1	4.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	11.8	17.7	16.1	11.4	17.3	15.6	11.1	16.9	48.0	18.2	46.5	17.5	45.2	17.2	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.9	13.5	12.4	DB	1.1	20.6	1.3	1.2	0.1	21.4	-0.6	22.2	-1.3	22.9	-2.3	23.8	(4)
(5)			WB	-0.7	18.4	1.2	1.1	-1.5	19.2	-2.2	19.8	-2.9	20.4	-3.8	21.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	7.6	7.1	7.5	9.0	11.1	13.5	15.2	15.7	15.1	13.2	11.0	8.6
	DBStd	3.52	2.16	1.96	1.70	1.51	1.38	1.36	1.38	0.97	1.30	1.74	1.89	2.30
	HDD10.0	344	77	82	77	37	5	0	0	0	0	2	12	52
	HDD18.3	2594	334	315	334	281	226	146	99	83	97	158	221	301
	CDD10.0	793	2	1	1	6	38	104	160	176	153	102	41	10
	CDD18.3	1	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.5	7.8	6.9	6.2	5.8	5.9	5.2	5.3	5.6	6.1	7.3	7.9	8.0
Precipitation	PrecAvg	1245	143	101	77	76	76	73	79	92	85	142	155	147
	PrecMax	1590	251	256	148	159	180	178	194	209	141	271	286	300
	PrecMin	984	5	27	25	18	11	21	18	6	14	47	78	25
	PrecStd	151	57	52	35	35	36	42	41	47	31	57	55	68
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	11.4	11.5	13.7	16.4	18.6	20.0	19.7	18.8	17.2	14.9	13.7
		MCWB	11.1	10.6	9.3	11.4	14.4	16.0	17.7	17.9	16.6	15.5	13.6	12.1
	2%	DB	11.4	10.7	10.6	12.2	14.6	17.2	18.9	18.2	18.0	16.4	14.4	13.1
		MCWB	10.3	9.9	9.0	10.5	12.9	15.1	17.1	16.5	16.1	14.9	13.1	11.3
	5%	DB	10.7	10.0	10.0	11.5	13.5	16.0	17.9	17.5	17.3	15.9	14.0	12.4
		MCWB	9.6	9.1	8.8	10.1	12.0	14.2	16.3	16.1	15.7	14.5	12.7	10.8
	10%	DB	10.2	9.5	9.5	10.9	12.8	15.2	17.1	17.1	16.8	15.4	13.4	11.6
		MCWB	9.1	8.5	8.4	9.6	11.5	13.7	15.8	15.8	15.4	14.0	12.0	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.1	10.0	11.7	14.7	16.5	18.1	18.6	17.1	15.9	14.1	12.7
		MCDB	11.7	11.3	10.5	13.0	15.8	18.0	19.5	19.6	18.0	16.6	14.5	13.3
	2%	WB	10.7	10.2	9.5	10.9	13.1	15.3	17.2	17.0	16.5	15.3	13.5	12.1
		MCDB	11.2	10.5	10.1	11.9	14.2	16.6	18.4	17.6	17.5	16.1	14.2	12.9
	5%	WB	10.0	9.3	9.1	10.3	12.4	14.6	16.6	16.6	16.1	14.8	12.9	11.2
		MCDB	10.5	9.8	9.7	11.1	13.2	15.7	17.7	17.2	17.0	15.6	13.7	12.1
	10%	WB	9.4	8.6	8.6	9.8	11.7	14.1	15.9	16.1	15.6	14.3	12.2	10.5
		MCDB	10.0	9.3	9.4	10.6	12.5	15.0	16.9	16.9	16.7	15.2	13.2	11.4
Mean Daily Temperature Range	5% DB	MDBR	2.3	2.1	2.2	2.1	2.0	2.0	1.7	1.7	1.9	2.2	2.1	2.4
		MCDBR	2.7	2.3	2.5	2.9	3.1	2.8	2.7	2.2	2.4	2.6	2.5	2.9
		MCWBR	3.4	2.5	2.5	2.3	2.5	2.4	2.1	2.1	2.3	2.8	3.1	3.7
	5% WB	MCDBR	2.6	2.2	2.4	2.6	2.9	2.6	2.5	2.1	2.1	2.5	2.5	2.9
		MCWBR	3.4	2.6	2.5	2.3	2.4	2.4	2.1	2.2	2.4	3.0	3.1	3.8
Clear-Sky Solar Irradiance	taub	0.332	0.344	0.371	0.387	0.394	0.384	0.388	0.371	0.365	0.353	0.330	0.322	
	taud	2.323	2.375	2.323	2.308	2.338	2.392	2.387	2.448	2.457	2.440	2.381	2.301	
	Ebn at Noon	650	766	814	847	858	869	859	856	817	746	651	588	
	Edh at Noon	63	80	101	116	118	113	112	100	88	74	59	55	
All-Sky Solar Radiation	RadAvg	0.71	1.45	2.69	4.26	5.30	5.75	5.20	4.36	3.18	1.76	0.87	0.53	
	RadStd	0.07	0.13	0.21	0.29	0.35	0.49	0.46	0.29	0.21	0.15	0.08	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page